

Nonlinear dynamics of two types of network with intelligent nodes

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In this paper, two types of network with intelligent nodes are studied. The first type has nodes that accumulate and delete information, while the second type has nodes that negotiate about a resource. It is established that the first network has multiple equilibria, and a condition for the existence of such equilibria is given. For the second network, it is shown how the negotiating strategy of the nodes influences the bandwidth required in the network.

1. Introduction

The work described in this paper is part of an ongoing study about stability in communications networks which contain nodes exhibiting some degree of intelligence. Instead of seeing the nodes as mere transmitters and receivers of bitstrings, the actual processing that is going on in the nodes is modelled. In this way the communications network with intelligent nodes becomes a network of nodes performing parallel distributed processing [1]. This makes it possible to analyse communications networks using neural network techniques. The main advantage of this approach is that results from neural networks provide information about global, emergent properties of the communications networks, for example the number of equilibria. The disadvantage is that the processing in the neurons is extremely simplified. This work develops more sophisticated neurons that come closer to modelling the intelligence of the nodes in a communications network. What is retained of the parallel distributed processing model is the massively parallel character, i.e. large groups of nodes have the same functionality.

2. Predicting bandwidth

Have you ever attempted to read a document on a popular US World Wide Web site from Europe? Even in the quiet European early morning hours, the transmission rate falls below a few bytes per second. At this speed, access to graphics, or interactive use becomes impossible. More bandwidth is needed, or the customers turn away from the Web site.

The notion of bandwidth arises from Shannon's paradigm of a communications channel [2]. Bandwidth is also a cost factor for the telecommunications operator. Although the required bandwidth of a communications channel has to be predicted, Shannon's paradigm may lead to a bad prediction in a World Wide Web environment. Indeed, Shannon's communications channel capacity was originally developed for communications between two users. Although extensions to multi-user systems have been found [3], the capacity in networks or distributed systems has not been well studied at all. Flows in networks have been studied [4], but the results are mostly in the form of algorithms, and they become cumbersome for networks with feedback — and feedback is everywhere in a World Wide Web with many hyperlinks. Capacity can be generalised for arbitrary graphs [5], but in doing so, one is far removed from Shannon's original paradigm. It may therefore be useful to look for other paradigms, other notions of information and capacity, that may lead to better predictions of the underlying bandwidth requirements.

The World Wide Web is an example of a distributed system, a distributed database in fact. How will it evolve? Currently it is in a growth stage. The number of nodes is increasing very fast. If all nodes had the same functionality, and were equally likely to communicate with each other, it would be easy to predict the increase in bandwidth needed from the increase in the number of nodes. However, the functionality of the nodes differs. Some sites are repositories of several gigabytes of data. Other sites consist

of users of these databases. Some users are satisfied with still images, others want to download video. This needs to be modelled somehow.

Consider the following event. Two databases are added to the network. One contains all the laws and regulations of the EC. The other contains videoclips. Assume that both databases contain 100 gigabytes of data, and are accessible at no charge. They are located in different sites. A network operator needs to predict the usage that will be made of the two databases. This usage needs to be predicted before the databases are operational, because, especially for the videoclip site, publicity will attract the user's attention for a few weeks, and if the communications are too slow, the users will stop connecting to it, and forget about the site. It is clear that very few users will connect to the EC site, and many, many more to the videoclip site. Using Shannon's notions of information, it is impossible to predict this from the contents of the database. What is needed is a characterisation of the site, and a poll among users of existing, similar sites, asking how often they connect to the site. It seems as if the job of predicting bandwidth has become more the job of the market researcher than of the engineer. The market researchers will model the preferences of the customers. In order to predict the stability of the network from these preferences, a new notion of information is necessary. Using this new notion, it will be shown in section 3 that the network settles in attractor states or equilibria, where the amount of information in the sites does not fluctuate much. As will be shown, the bandwidth will influence the preferences, and these will influence the stability.

A network of intelligent nodes cannot exist outside of economic reality. If intelligent nodes contain information that is of value to other nodes, they will try to predict this value, so that they can sell the information to the other nodes. This leads on to modelling a network of economic agents. Section 4 studies agents that adapt the value of their time via a differential law. This situates the study of networks with intelligent nodes within the framework of general equilibrium theory [6] and differential games [7]. The contribution here is to show that this framework can be used to predict bandwidth in a network of agents negotiating about the value of their time.

3. Attractors in a communications network

In this section, a communications network is modelled by an artificial neural network with a special type of neuron. It will be shown that this can model some essential constituents of intelligence, namely accumulation, rejection, and saturation of information. Once the parameters in the model are defined, it becomes possible to apply neural network techniques to study the stability of the system. Existing neural network results cannot be simply transferred, because the neurons are modified in this model.

3.1 Weights and information exchange

The information that a communications node has is represented by a number between 0 and 1. This number is the state of the node. If a node is saturated with information, its state is 1. If it has no information, its state is 0. Every node has its own scaling factor, so that saturation with information means that the state of the neuron is 1. A person with a multimedia workstation will have more information when he is saturated than a farmer in Africa, but both will be in a state 1 if they are saturated with information. The state of a node will change in time.

Information is transferred with a certain efficiency between nodes. This efficiency may depend on the quality of the communications line between the nodes, as in standard communications theory. In this respect, the efficiency is comparable to the capacity of a communications channel. In this model, however, the efficiency can also depend on the nature of the information that is communicated. A database may contain a large amount of information that is only of minor use to a communications node interrogating the database. In this case, the amount of information in the database will be close to 1, but the interrogating node will only require a small amount of the available information.

In general, this can be formulated as follows. If node j has an amount of information x_j , and communicates with node i with efficiency T_{ij} , the amount of information that is added to x_i is $T_{ij}x_j$.

The experimental measurement of the T_{ij} would be conducted in the same way as an opinion poll. At a particular moment in time, every node will be asked how much information it contains, and, for each of the other nodes it connects to, how much information it gains every time it connects to it. These quantities are equal to x_j and $T_{ij}x_j$ respectively. The quantity T_{ij} will then be equal to the information gain $T_{ij}x_j$ (supplied by node i), divided by the information x_j present in node j (supplied by node j). The efficiency may depend on the bandwidth of the communication line, but also on non-technological factors, e.g. the interest that a user has in particular data. Assume now that there are n nodes, and that node i gets information from each of them. It will be assumed that the information is superimposed linearly. The efficiencies T_{ij} , $i = 1, \dots, n$, $j = 1, \dots, n$, will be scaled between 0 and 1. If $T_{ij} = 0$, there is no information flow from j to i . If $T_{ij} = 1$, all information present in node j flows to node i if there is communication from j to i . Note that $T_{ij} \geq 0$.

Whereas small amounts of information can be superposed linearly, once the information of a node is close to 1, it saturates in such a way that for a large inflow of information, the information content of a node approaches 1

asymptotically. This is obtained by applying a nonlinear saturating function $(1 + \tanh x)/2$. This function is called σ and the input is replaced by:

$$\sigma\left(\sum_{j=1}^n T_{ij}x_j\right)$$

The exact form of this function would have to be determined from psychological experiments.

Any forgetting of information is assumed to be proportional to the amount of information already present in the node. If a node is close to saturation, one needs to flush some old information. This is modelled by subtracting a positive function $f(x_i)$ from the input to x_i . The concept of forgetting is intuitively motivated by the fact that while information gradually builds up as time proceeds, some old information leaks away. This leaking away may be actual forgetting, as has been observed in short-term human memory, but it can also mean the actual purging of data, for example when old files are deleted.

The input to node i from within the communications network is:

$$\sum_j T_{ij}x_j(t) - f(x_i(t))$$

The external input to node i is denoted by the single variable θ_i . This external input of information comes from the databases to which the communications node i connects. The equation for the evolution in time is:

$$x_i(t+1) = \sigma\left(\sum_{j=1}^n T_{ij}x_j(t) - f(x_i(t)) + \theta_i\right) \quad i = 1, \dots, n. \quad \dots(1)$$

The quantity θ_i is the total amount of information node i gets from all the databases to which it connects, for one time unit. This quantity is assumed to be independent of time, because the time-scale on which it changes is larger than the time-scale on which the amount of information in a node changes. To rephrase this in a World Wide Web context, although the number of sites may grow exponentially, the amount of information a user gets from them increases much more slowly.

If the amount of information x_i is below a threshold a , nothing should be forgotten, and therefore $f(x_i) = 0$ if $x_i < a$. If x_i is greater than a , the amount of information should be decreased, regardless of the inflow of information to node i .

This will happen if f equals the maximum input to the node. Finally, there is the following choice for the function f .

$$f(x_i) = \begin{cases} 0, & x_i < a \\ b\left(\sum_{j=1}^n T_{ij} + \theta_i\right) & x_i \geq a \end{cases} \quad \dots (2)$$

The factor b , which is positive, models the strength of the forget action.

Some neural network models have forgetful neurons that are leaky integrators. In this model, where the state of a neuron represents an amount of information, this would mean that information leaks away according to a certain time constant. This model was not chosen, because the forgetting of information needs to be a function of the amount of information present, and not of time only. Humans may slowly and uniformly forget information, but when a disk is full, information will be purged in blocks, it will not gradually fade. To some extent the latter process also occurs in humans, when they lose interest in a whole subject area.

3.2 Equilibria

The set of n coupled difference equations (1) also represents the dynamics of a neural network, apart from the forget function f , which introduces a self-feedback to the neurons. The network is completely connected, i.e. every neuron can be connected to every other neuron. In neural network terms, the θ_i , $i = 1, \dots, n$, are the external inputs to the n neurons.

There is an important difference with the Hopfield neural network [8] in the way the weights in equation (1) are determined. In the normal use of a neural network, the user specifies some equilibria (also called patterns, fundamental memories, or attractors), and calculates the matrix T from these equilibria with a rule, e.g. sum-of-outerproducts [8]. In this model, the elements of T are efficiencies of communication links, or preferences of users. They have to be determined by statistical surveys of the users. They depend on what media the network databases have on offer, and what information other users in the network possess.

The model of equation (1) has desirable properties for a dynamic analysis, e.g. the external inputs θ_i are independent of time. It has also been assumed that the weight matrix T is symmetric, or that the communication efficiency from node i to j is the same as that from node j to i . Despite this, there does not appear to be a Lyapunov function for equation (1), and the negative feedback $-f(x_i(t))$ is responsible for this. This affects directly the analysis of the equilibria. For the

simulations, it will be assumed that an equilibrium is reached when the information in the nodes does not change any more within the accuracy of the computer.

The communications network is exchanging information in an asynchronous mode. The simulations are performed asynchronously as well, selecting neurons at random for update. The next step is to vary the parameters in equation (1). Two parameters are obvious, they are the forget threshold a and the forget rate b , introduced via the forget function f in equation (2). The third parameter is related to the external input θ_i to the neurons. It will be assumed that θ_i is a random variable, uniformly distributed between 0 and $c > 0$. This defines the third parameter, c . Finally, the T_{ij} were randomly distributed between 0 and 1, and T was chosen to be a symmetric matrix.

How many equilibria are there? This is illustrated in Fig 1. If $c = 1$, it is found that for $b < 0.3$, there is only one equilibrium, with all nodes saturated at 1. For $0.3 < b < 0.8$, there are many equilibria, roughly as many as there are neurons. This is not surprising in a neural network. For $b > 0.8$, Fig 1 shows multiple equilibria, but closer inspection of the simulation results shows that most of the nodes are in state 0.

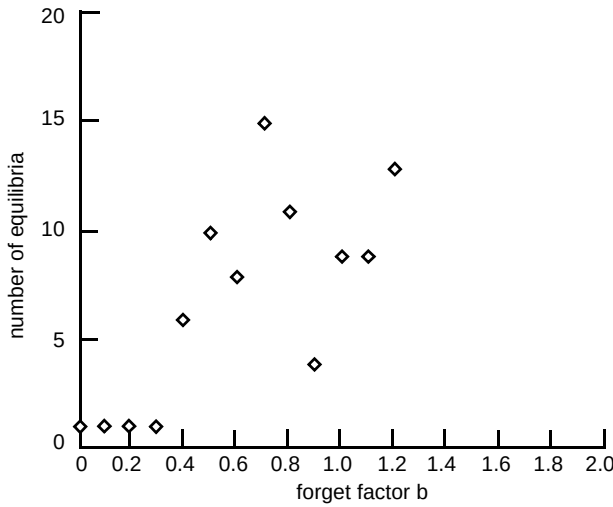


Fig 1 The number of equilibria, for 15 neurons, for various forget factors b . For $b \leq 0.3$, there is only one equilibrium, with all nodes in saturation. For $b \geq 0.8$, the equilibria are zero nearly everywhere.

The same region $0.3 < b < 0.8$ of multiple equilibria was also found for $n = 20$ nodes, and for thresholds $a = 0.1, 0.5, 0.9$. Varying c , most nodes became saturated for $c \geq 3$.

In summary, it has been established that, for many values of the parameters a , b , and c in the network, multiple equilibria exist. The only limitation was that the number of neurons is small. This limitation can be circumvented by a theoretical analysis.

As time evolves, the state $x_i(t)$ of neuron i takes on values between 0 and 1. As criterion for stability of the state of neuron i , the following will now be used: if $x_i(t) > 0.5$ at a certain time t , $x_i(t)$ has to stay larger than 0.5. Alternatively, if $x_i(t) < 0.5$, it has to stay smaller than 0.5. This means that $2x_i(t+1) - 1$ and $2x_i(t) - 1$ have to have the same sign, or:

$$[2x_i(t+1) - 1][2x_i(t) - 1] \geq 0 \quad \text{..... (3)}$$

for all t .

This stability criterion is based on an analysis of the transients. It was found experimentally that for most transients, the x_i change by more than 0.67. Therefore, some stability will occur if the x_i do not cross the 0.5 value for a long time. If one wants a stricter criterion, e.g. $x_i < p$ or $x_i > 1 - p$, condition (3) becomes:

$$[2x_i(t+1) - 1][2x_i(t) - 1] \geq (2p - 1)^2 \quad \text{..... (4)}$$

for all t . This criterion guarantees equilibria where the nodes are assumed to be either saturated, $x_i = 1$, or empty, $x_i = 0$.

Using equation (1), condition (3) becomes:

$$\sum_{j=1}^n T_{ij}x_j(2x_i - 1) - f(x_i)(2x_i - 1) + \theta_i(2x_i - 1) \geq 0 \quad \text{.....(5)}$$

The sum over j in the left-hand side expresses how the other nodes disturb the stability of node i (remember that $T_{ii} = 0$) — this sum is the ‘noise term’. The part $-f(x_i)(2x_i - 1) + \theta_i(2x_i - 1)$ in the left-hand side contains only x_i — it is the ‘signal term’.

For communications networks with a large number of nodes, and after an extensive mathematical analysis, several conclusions can be drawn from this equation, conclusions about how the parameters a , b and c influence the stability. For example, one of the calculations implies that the condition (5) can only hold if:

$$\lim_{n \rightarrow \infty} -\frac{27}{\sqrt{2}}(2a - 1)b = \infty \quad \text{.....(6)}$$

This leads to: ‘For a communications network with a large number of nodes to have stable equilibria, it is necessary that the forget threshold a is smaller than 0.5 and that the strength of the forget action b grows with the number of nodes’.

Because theoretical results, such as this, are asymptotical for the number of nodes going to infinity, they complement the simulations, which are for a small number of neurons.

3.3 Conclusion

A network of communication nodes has been analysed using a common-sense concept of information. If the efficiency of the communication between the nodes can be estimated (parameters T_{ij}), and if it has been determined how the nodes forget information (parameters a and b), then it was possible to establish that multiple equilibria can exist in the network. In such an equilibrium state, the information in the nodes does not change over time. This is what is observed in many real communications networks — apart from statistical fluctuations, many users possess an amount of information that does not change greatly over time.

However, in the real world, only one equilibrium is observed. If the network operator knows the other equilibria, they can influence the free parameters T_{ij} , for example by providing extra bandwidth, so that the network settles in another equilibrium that is more profitable for the operator:

extra bandwidth between i and $j \Rightarrow$ larger $T_{ij} \Rightarrow$ other equilibria.

How exactly one can steer the network from one local minimum (equilibrium) to another, is described in Buhmann and Schulten [9].

4. Intelligent nodes

4.1 Definition

As has been said in the introduction, there is not much intelligence in a network where some nodes give away information free, and the others accumulate (and forget) this information. In a more realistic world, the nodes consider their information as a ‘resource’ that has a value associated with it. If the resource is information, then communication consists of the exchange of information between the nodes, and this has been analysed in section 3. If the resource is goods, or time, or personnel, the nodes will still communicate, not to exchange the goods, but to negotiate about the goods. In a network of intelligent nodes, each representing a resource, arguably, this negotiation will become so important that the communications it generates could take up nearly as much bandwidth as the exchange of information. Important negotiations are not confined to big contracts. It will be shown in an example that even a relatively inexpen-

sive resource such as somebody’s time can generate a lot of negotiation if the person is extensively ‘connected’ within the organisation.

The basic structure for every resource (every node) will be its location in time, its value as a function of time, and its description as a function of time. If there are n resources, the location of resource i at time t , will be denoted by $l_i(t)$, the value of resource i at time t by $v_i(t)$, and its description by $d_i(t)$. It is assumed that the locations are numbered, so that $l_i(t)$ is a number. The value $v_i(t)$ is *de facto* a number. The description $d_i(t)$ is a file that may contain ASCII text, and drawings or pictures. It will be argued in another paper that this provides the data structure necessary to develop intelligence, in the sense of Piaget [10].

Consider the example of a meeting. Employee i has to attend the meeting between the times t_1 and t_2 . Say the room number is location 3. Then:

$$l_i(t) = 3, \quad t_1 < t < t_2$$

The employee knows that the first five minutes of the meeting are seldom interesting, because some participants usually are late. The value function could be (assuming time is measured in minutes):

$$v_i(t) = \begin{cases} 0, & t_1 < t < t_1 + 5 \\ 4, & t_1 + 5 < t < t_2 \end{cases}$$

indicating that the employee values his time at 4 units per minute during the effective part of the meeting.

The function l could change, if it is decided to change the location of the meeting, or the start or finishing time. The function v could change if for example it becomes known that somebody will participate who is usually very late, and without whom the meeting cannot start.

The meeting is in the future for $t < t_1$, and d will then contain only the agenda. During the meeting, minutes are taken, and directly entered into a file. For times $t > t_2$, the description d contains the minutes of the meeting.

4.2 Stability considerations

In general, a model such as the one that has just been described, will show a ‘dynamical behaviour’. Every node (resource) has time-dependent functions associated with it. These functions will change, dependent on the state of the other nodes. For example, the location of a meeting can be changed. This will imply changes of $l_i(t)$ in several nodes; or an important person can decide to attend the meeting, so

that its value $v_i(t)$ becomes larger for the other participants. Before the meeting, the descriptor $d_i(t)$ contains the agenda for the meeting, after the meeting has taken place, the descriptor contains the minutes of the meeting.

In order to illustrate how this dynamical behaviour can be analysed, let us consider a simple system where location and descriptor stay the same, and only the value $v_i(t)$ changes. Every node i changes its $v_i(t)$ at a certain moment t in time, depending on the v s of the other nodes according to a strategy. A general strategy for change of $v_i(t)$ is:

$$\frac{dv_i}{dt} = f_i(v_1, v_2, \dots, v_n) \quad \dots(7)$$

in a network of n nodes. This simply indicates that the value of a resource associated with the node i changes in time as a function f_i of the value of the other nodes.

This is a simple differential formulation of negotiation, considered as a game [7, 11]. If the change in the values is governed by a system of coupled differential equations of the form of equation (7), the nodes will change their values simultaneously. Some nodes may be slower than others, and this can be modelled by introducing delays in the values of $v_1, v_2, \dots, v_n$ used in the functions f_i .

Often the resource is a person's time. This time is different (has a different clock) from the time t in equation (7). This will be illustrated later in an example.

The update equation (7) to the value is performed by all nodes concurrently. Will they settle (reach an agreement)? This will depend on whether the dynamical system:

$$\frac{dv_i}{dt} = f_i(v_1, v_2, \dots, v_n), \quad i = 1, \dots, n \quad \dots(8)$$

reaches an equilibrium. In vector notation, the system can be denoted by $\dot{\mathbf{v}} = \mathbf{f}(\mathbf{v})$. This can be linearised as:

$$\dot{\mathbf{v}} = \mathbf{A}\mathbf{v} + O(\mathbf{v}^2) \quad \dots(9)$$

The nodes will settle if the eigenvalues of the matrix $\mathbf{A}$ have strictly negative real parts. If this is the case, and if

$$-\alpha > \max(\Re \lambda_1, \Re \lambda_2, \dots, \Re \lambda_n) \quad \dots(10)$$

is chosen, then the settlement is exponentially stable [12]:

$$\|\mathbf{v}(t)\| \leq \|\mathbf{v}(t_0)\| e^{-\frac{\alpha(t-t_0)}{2}} \quad \dots(11)$$

The symbol $\Re$ denotes the real part of the eigenvalue.

If the nodes favour their own opinion above the others in their strategy, then the matrix $\mathbf{A}$ will be diagonally dominant — the settlement will be fast. If the nodes attach a lot of importance to the $v_i(t)$ of the other nodes, the settlement will be democratic, but slow. All this is described by the value of α .

Example — assume a firm operates from two sites, e.g. London and Martlesham. In each site there are 1000 intelligent nodes. Every node values its resource dependent on the value of the resources of the other nodes. If one node changes the value of its resource for a 30-minute period, what extra bandwidth is needed on the trunk line between the two sites, so that all nodes can re-value their own resource?

Answer — let it be assumed that every node keeps track of the value $v_i(t)$ of its resource at points in time one minute apart. If one node changes its $v_i(t)$ over a 30-minute interval, 30 values of $v_i(t)$ have to be renegotiated over the network. How long will this take? Consider the renegotiation for one value in time t_0 , for all nodes. The strategy f_i for every node has to be known. This is a system identification problem (see, for example, Gene et al [13]). Now calculate the eigenvalues from $\mathbf{A}$ in equation (9). If the eigenvalues all have strictly negative real parts, the global strategy of all nodes will lead to a consensus over $v_i(t_0)$, $i = 1, \dots, n$. If not, a special study can be made to discover the nodes that have a strategy that prevents a consensus. Calculate α from equation (10).

For this one moment t_0 in time, every node has to recalculate its $v_i(t_0)$. The nodes start by making an initial guess for $v_i(t_0)$. A consensus is reached if the initial guess has been improved by 99%. From equation (11) it can be calculated that the number of calculation steps needed for 99% improvement is $-(2 \ln 0.01)/\alpha$, or approximately $9.2/\alpha$. One such step involves the reading of 2000 values of $v_i(t_0)$, 1000 of which have to be transmitted over the trunk line, and also involves the calculation of f_i . Assume that the calculation time is negligible compared with the time needed to transmit the data. (This will be the case if f_i is a straightforward function, and does not require numerical algorithms or access to databases.) If x is the extra transmission speed in bits per second (bit/s) required on the trunk line, and if each $v_i(t_0)$ requires 32 bits, one node

requires $32\,000/x$ sec to update its value of $v_i(t_0)$ according to its strategy.

The thousand updated values of Martlesham have to be transmitted only once to London, where they can be broadcast to the 1000 London nodes over a LAN. The same for the transfer of the 1000 updated London values to Martlesham. This means that $64\,000/x$ sec is needed to update $v_i(t_0)$ in all 2000 nodes, each according to their own strategy. This has to be done $9.2/\alpha$ times before a consensus is reached. This whole process has to be repeated 30 times, to cover the 30-minute interval over which the one node had changed its $v_i(t)$. This means that the whole update process takes:

$$\frac{64000}{x} \cdot \frac{9.2}{\alpha} \cdot 30 \text{ sec} \quad \dots(12)$$

Such an update process has to be finished within a minute, because the resources are valued every minute, and a node may want to use its $v_i(t)$ function to calculate something else, a budget forecast for example. It will also be assumed that during this one minute interval, only 10 nodes out of the 2000 change something to their $v_i(t)$. The exact number could be measured, e.g. from the frequency with which people change something in their diaries. Converting minutes into seconds, the following inequality is found for the extra transmission speed x in bit/s:

$$\frac{64000}{x} \cdot \frac{9.2}{\alpha} \cdot 30 \cdot 10 \leq 60 \quad \dots(13)$$

or

$$x \geq \frac{2.9}{\alpha} \text{ Mbit/s} \quad \dots(14)$$

This shows that the eigenvalues of the strategy with which the intelligent nodes value their resource determines the transmission speed required for the communication between the nodes.

5. Conclusions

This paper has shown that it is possible to analyse the dynamical behaviour of a network of intelligent nodes. If the nodes reach a consensus, it is possible to predict what communications bandwidth this will require. The nodes were negotiating the value of a resource, based on the values of other resources. A more complete negotiation would also take into account the location and descriptors of the other resources. That this is a general model of an intelligent node in a network will be argued elsewhere. More work is required on the analysis of the dynamical behaviour of the network, especially to determine when a consensus is reached. Some nodes can have an anomalous strategy that

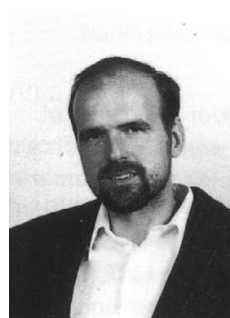
prevents a consensus being reached. The network will then become unstable, generating large amounts of communication between the nodes. Work on this problem is currently being undertaken.

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References

- 1 Rumelhard D E and McClelland J L: 'Parallel Distributed Processing: Explorations in the Microstructure of Cognition', 1, MIT Press, Cambridge, Massachusetts (1986).
- 2 van Lint J H: 'Introduction to Coding Theory', Springer-Verlag, New York (1982).
- 3 van der Meulen E C: 'A survey of multi-way channels in information theory', IEEE Transactions on Information Theory, IT-23, No 1, pp 1—37 (1977).
- 4 Ford L R and Fulkerson D R: 'Flows in Networks', Princeton University Press, Princeton (1962).
- 5 Gargano L, Korner J and Vaccaro U: 'On the capacity of Boolean graph formulas', Graphs and Combinatorics, 11, No 1, pp 29—48 (1995).
- 6 Debreu G: 'Theory of value', John Wiley, New York (1959).
- 7 Basar T and Olsder G J: 'Dynamic Non-co-operative Game Theory', Academic Press, New York (1982).
- 8 Hertz J, Krogh A and Palmer R G: 'Introduction to the Theory of Neural Computation', Santa Fe Institute Studies in the Sciences of Complexity, Addison-Wesley, Redwood City, California (1991).
- 9 Buhmann J and Schulten K: 'Noise driven temporal association in neural networks', Europhysics Letters, 4, No 10, pp 1205—1209 (1987).
- 10 Piaget J: 'Six etudes de psychologie', Folio essais, Denoel, Paris (1964).
- 11 Nicolis J S: 'Dynamics of Hierarchical Systems', Springer Verlag, Berlin (1986).
- 12 Abraham R and Marsden J E: 'Foundations of Mechanics', Benjamin, New York (1967).
- 13 Gene F, Franklin J, Powell D and Workman M L: 'Digital Control of Dynamic Systems', Addison-Wesley, Reading, Massachusetts (1990).



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